Ratnakar Gadi

Postdoctoral Research Associate
Environmental Science and Engineering, California Institute of Technology
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Research Interests

Ice–ocean interactions across scales; seawater exchange and basal melting in ice grounding zones; tidal flexure and sub-ice-shelf circulation; residual overturning, eddy transport, and water-mass transformation; integration of numerical models with satellite and in situ observations.

Education

Ph.D., Earth System Science 2021–2025
University of California, Irvine (UCI). Advisor: Eric Rignot.
Thesis: *Comprehensive analysis of the melt regime in Ice Grounding Zones (IGZ).*

M.Sc., Maritime Technology 2015–2018
Delft University of Technology, The Netherlands. Advisor: Rene Huijsmans.
Thesis: *Efficient methodology of roll prediction of two-dimensional bodies in nonlinear flows.*

B.E., Naval Architecture and Marine Engineering 2010–2014
Andhra University College of Engineering, India. Advisor: I.R.N. Niranjan Kumar.
Thesis: *Preliminary design of a 112257.6 tonne aircraft carrier with a payload of 675 tonnes.*

Academic and Research Appointments

Postdoctoral Research Associate, Caltech 2025–present
Environmental Science and Engineering, Division of Geological and Planetary Sciences. Advisor: Andrew Thompson.

Project Associate, Indian Institute of Science 2019–2021
Department of Computational and Data Science, Bangalore, India.

Junior Project Assistant, IIT Kharagpur 2018–2019
Indian Institute of Technology Kharagpur, India.

Research Intern, TU Delft Summer 2016
Faculty of Maritime Technology, The Netherlands.

Peer-Reviewed Publications

- **Gadi, R.**, Rignot, E., Menemenlis, D., and Scheuchl, B. (2025). Contrasting melt regime in the grounding zone of Thwaites Glacier, West Antarctica. *PNAS*, 122(48), e2512626122. **Recognition:** "In This Issue" highlight and press tip.
- Rignot, E., Chauche, N., **Gadi, R.**, Ehrenfeucht, S., and Ciraci, E. (2025). Grounding line ROV survey of the ice shelf cavity of Petermann Glacier, Greenland. *Geophysical Research Letters.*
- Chen, H., Rignot, E., Scheuchl, B., Milillo, P., Morlighem, M., **Gadi, R.**, et al. (2025). Rapid retreat of Berry Glacier, West Antarctica. *Nature Communications.* **Recognition:** feature highlight.
- **Gadi, R.**, Rignot, E., and Menemenlis, D. (2023). Modeling ice melt rates from seawater intrusions in the grounding zone of Petermann Gletscher. *Geophysical Research Letters.* **Recognition:** EOS Research Spotlight and dedicated GRL commentary.

- **Gadi, R.**, Vinayachandran, P.N., and Subramani, D.N. (2021). Data-driven feature-oriented modeling of Southwest Monsoon Current. *Ocean Modelling*.

Manuscripts in Preparation

- **Gadi, R.**, Thompson, A., Menemenlis, D., and Rignot, E. (2026). Overturning and basal melt in high-resolution simulations of tidal flexure beneath an idealized ice shelf. *In Development (submission forthcoming)*.
- Charrasin, R., Rignot, E., Millan, R., Gimenes, L., Chen, H., **Gadi, R.**, Chauche, N., and Nicholls, K. (2026). Spatial and temporal variability of basal melting at the grounding zone of Roi Baudouin Ice Shelf, Queen Maud Land, East Antarctica. *In Development (submission forthcoming)*.
- Chen, H., Rignot, E., Scheuchl, B., **Gadi, R.**, Charrasin, R., and Poinelli, M. (2026). Two decades of Antarctic ice-shelf basal melting across shifting ice grounding zones. *Submitted*.

Conference Proceedings

- **Gadi, R.**, Huijmans, R.H.M., Akkerman, I., and Veer, R.V. (2018). Efficient methodology of roll load prediction on 2D bodies in nonlinear flows. *Proceedings of the International Workshop on Water Waves and Floating Bodies*.

Research Software and Open Science

Developed and publicly released research code on Zenodo:

- **ICESat-2 melt-rate estimation:** zenodo.org/records/12785929.
- **TanDEM-X melt-rate estimation:** zenodo.org/records/12658529.
- **Operation IceBridge data processing:** zenodo.org/records/12659151.

Methods and Technical Expertise

- **Ocean modeling:** source-level MITgcm development for two- and three-dimensional grounding-zone and ice-shelf cavities; time-dependent ice-load forcing, grounding-line migration, shelf-ice remeshing; idealized and realistic cavity simulations.
- **Transport diagnostics:** residual overturning, water-mass transformation, eddy heat transport, diapycnal transport, and heat and volume budgets for moving-boundary simulations.
- **Satellite and airborne observations:** ICESat-2 ATL14/15 melt-rate estimation; TanDEM-X DEM time series for Lagrangian melt-rate estimation; Operation IceBridge ATM LiDAR and MCoRDS radar processing to investigate static bending zones.
- **In situ observations:** ApRES processing and model comparison; seismic hammer and Geode data processing to identify ice-base and seabed reflections; ROV observations of ice-shelf base and bed elevation.
- **Programming:** MATLAB (advanced), Fortran (advanced), Python (intermediate/advanced).

Field Experience

- **Petermann Gletscher, Greenland** (May 2023): ROV measurements of ice-shelf base and bed elevation, along with CTD cast collection.

- **Roi Baudouin Ice Shelf, East Antarctica** (January 2025): seismic measurements of ice thickness and seabed depth, and ApRES measurements of ice thickness and basal melting. Assisted with automatic weather station (AWS) installation and deployment of a GNSS station for NISAR calibration and validation.

Honors and Awards

- Outstanding Contribution Award, Earth System Science Department, UCI (2023–2024).
- Best Teaching Assistant Award, Earth System Science Department, UCI (Fall 2023).
- Runner-up, TUCK Fellowship, International Workshop on Water Waves and Floating Bodies (2018), for "Efficient methodology of roll load prediction on 2D bodies in nonlinear flows."
- Most Dynamic Student Award (2013–2014), Department of Naval Architecture and Marine Engineering, Andhra University, India. Recognized for contributions to planning and organizing the departmental technical festival and outstanding undergraduate thesis work.

Publication highlights and commentary recognition are listed alongside the corresponding papers.

Leadership and Professional Service

- **IceNode science team member** (2026–present): assisted with device pressure testing and participated in science planning meetings with Andy Thompson, Tim Stanton, and Paul Glick for a planned November 2026 deployment at Filchner–Ronne Ice Shelf, Antarctica, in collaboration with the British Antarctic Survey.
- **CalGFD organizing committee and session chair** (2025, 2026): selected session abstracts with co-organizers, coordinated with presenters, chaired sessions, and facilitated scientific discussion. Designed the meeting webpage and supported poster-session setup in 2026.
- **Earth Day organization, UCI ESS:** worked with the department chair and communications lead to bring Earth Day activities back into the department.
- **Graduate funding communication, UCI ESS:** communicated concerns between faculty and graduate students during changes to summer-funding arrangements following unionization; worked toward arrangements to prevent students from going unsupported.

Teaching and Mentoring

- **Teaching assistant, UCI ESS:** recognized with the Fall 2023 Best Teaching Assistant Award. Encouraged independent problem solving and discussion of alternative approaches in an environment where students could express ideas confidently.
- Mentored a high-school student in research at Caltech (2025).
- Participated in the IARPC mentor–mentee program (2025, 2026).
- Advised UCI undergraduate students on career choices, drawing on experience across engineering and Earth system science.

Media and Public Engagement

- Featured in the documentary *La Lutte des glaces*.

- Featured in UC Irvine's "An academic first in Antarctica" (2025), covering the Antarctic expedition and dissertation defense at Princess Elisabeth Antarctica station and connecting the expedition to an international polar research and education initiative.

Oral Presentations

- **Gadi, R.**, Thompson, A., Menemenlis, D., and Drake, H. (2026). Coupling ice-shelf melt rates and cavity circulations: A water mass transformation perspective. IGS Ice-sheet/Ocean Interactions meeting, SIO.
- **Gadi, R.**, Rignot, E., Menemenlis, D., and Scheuchl, B. (2024). ITGC MELT: Contrasting melt regime in the grounding zone of Thwaites Glacier, West Antarctica. ITGC meeting.
- **Gadi, R.**, Rignot, E., and Menemenlis, D. (2023). On the impact of seawater intrusions on basal melt rates in the Ice grounding zone. AGU.
- **Gadi, R.**, Rignot, E., Menemenlis, D., Ciraci, E., Scheuchl, B., Tolpekin, V., and Wollersheim, M. (2023). On the impact of seawater intrusions on basal melt rates in the grounding zone of Thwaites Glacier, West Antarctica. WAIS conference.

Poster Presentations

- **Gadi, R.**, Thompson, A., Menemenlis, D., and Rignot, E. (2026). Tidal Control of Ice Shelf Melt Rates in the Grounding Zone: An Idealized Modeling Study. CalGFD, UCLA.
- **Gadi, R.**, Thompson, A., Menemenlis, D., and Rignot, E. (2025). Modeling of tidal-scale melt variations in the Ice Grounding Zone of idealized ice-shelves. AGU.
- **Gadi, R.**, Thompson, A., Menemenlis, D., and Rignot, E. (2025). Tidal Control of Ice Shelf Melt Rates in the Grounding Zone: An Idealized Modeling Study. CalGFD, UCSB.
- **Gadi, R.**, Rignot, E., and Menemenlis, D. (2024). Melt regime in the Ice Grounding Zone of Petermann Glacier, Greenland. AGU.

Other Co-Authored Conference Contributions

- Scheuchl, B., Rignot, E., Herreid, S., Barre, J.B., Poinelli, M., **Gadi, R.**, and Shamsian, S. (2025). Using Commercial Spaceborne SAR data for Ice Sheet Research. AGU, oral presentation.
- Rignot, E., Charrassin, R., Chauche, N., and **Gadi, R.** (2025). Ice melt processes on Roi Baudouin Ice Shelf, Queen Maud Land, East Antarctica. AGU, poster.
- Chen, H., Rignot, E., Scheuchl, B., and **Gadi, R.** (2025). Multi-year Changes in Antarctic Ice-Shelf Basal Melting. AGU, poster.